

CLASS 15+16: LEARNING RESOURCES

Lists & Conditionals	useEffect & APIs	■ AI Tools
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01 Official React Documentation

Rendering Lists — react.dev

<https://react.dev/learn/rendering-lists>

The official guide to `.map()` in JSX, keys, and why they matter. Covers the rules for keys and common mistakes. Read this before starting the homework. Short and very clear.

Conditional Rendering — react.dev

<https://react.dev/learn/conditional-rendering>

Covers all three patterns: `&&`, ternary, and early returns. Shows real examples of each. Read side by side with the code samples from class.

Synchronizing with Effects — react.dev (useEffect)

<https://react.dev/learn/synchronizing-with-effects>

The official `useEffect` guide. Covers the dependency array in detail, cleanup functions, and how to think about effects. This is the definitive reference.

You Might Not Need an Effect — react.dev

<https://react.dev/learn/you-might-not-need-an-effect>

Important read: covers the cases where people overuse `useEffect`. Helps you understand when `fetch` belongs inside `useEffect` and when it belongs in an event handler.

02 ■ AI Coding Tools — Setup & Usage

GitHub Copilot — Official Setup

<https://docs.github.com/en/copilot/quickstart>

Step-by-step setup guide for GitHub Copilot in VS Code. Covers account setup, extension installation, and your first suggestions. Follow this if you had trouble during class.

GitHub Copilot — Student Plan (Free)

<https://education.github.com/pack>

If you are a student, GitHub offers Copilot for free through the GitHub Student Developer Pack. Apply with your student email. Takes 1-3 days to verify.

Claude — Anthropic (Best for debugging)

<https://claude.ai>

Free tier available. Use for: pasting errors and getting explanations, code review, understanding concepts with plain-English analogies. Best AI for "why doesn't this work?" questions.

ChatGPT — OpenAI

<https://chat.openai.com>

Free tier available. Good for generating starter code, step-by-step explanations, and "how do I..." questions. Also useful for writing comments and documentation.

Prompt Engineering Guide — Learn AI Prompting

<https://www.promptingguide.ai>

Learn how to write better prompts. The better your prompt, the better AI's answer. Read the "Basic Prompting" and "Few-Shot Prompting" sections — they apply directly to coding.

03 YouTube Tutorials

The Net Ninja — React Tutorial (useEffect & Fetch)

Search YouTube: "Net Ninja React useEffect fetch data tutorial"

A focused video on fetching data with useEffect. Shows loading states, error handling, and how to display the results. Directly covers what was taught in class.

Web Dev Simplified — useEffect Explained

Search YouTube: "Web Dev Simplified useEffect explained"

Kyle Cook's clear explanation of useEffect, when to use it, and the dependency array. Watch this if you want a second explanation of what was covered in class.

Traversy Media — Async/Await & Fetch API

Search YouTube: "Traversy Media async await fetch API"

Solid foundation in fetch() and async/await — the JavaScript skills that power data fetching in React.

04 Interactive Practice

React.dev Interactive Tutorial — Describing the UI

<https://react.dev/learn/describing-the-ui>

The official React tutorial has interactive exercises for rendering lists and conditional rendering. These appear directly in the browser — no setup needed.

CodeSandbox — React Template

<https://codesandbox.io/s/new>

Build and test React components in the browser. Use this if your local setup has issues or to quickly test an AI suggestion before adding it to your project.

05 Quick Reference: Common Questions

Q: My list is not rendering. What do I check first?

1. Is your array actually populated? `console.log(yourArray)` before the return. 2. Is it an array or undefined? If undefined, you'll get "map is not a function." 3. Does every item have a unique key? Check the console for key warnings.

Q: What is the difference between `&&` and ternary?

"`&&`" shows something or nothing. If the condition is false, nothing renders. Ternary always renders one of two things. Use `&&` when you only want to show a badge if liked. Use ternary when you want to show "liked" or "not liked".

Q: Why does my `useEffect` run twice in development?

React 18 runs `useEffect` twice in strict mode (development only) to help detect bugs. This is expected. Your production build will not do this. If this causes problems, it usually means your fetch function needs cleanup.

Q: My fetch is failing. What do I check?

1. Open the URL in a browser tab — does it return data? 2. Check for typos in the URL (most common cause). 3. Open DevTools Network tab and look at the failed request for more detail. 4. Some APIs block browser requests (CORS) — check the API documentation.

Q: Can I really use AI for my homework?

Yes — and you should! But: try the problem yourself for at least 10 minutes first. When AI gives you code, read it and understand it before using it. If someone asks you to explain your code, you should be able to. AI is a learning accelerator, not a learning replacement.

Q: My `useEffect` runs in an infinite loop. Why?

You are either: (1) missing the dependency array entirely, or (2) watching a state variable that changes inside the effect. Fix: add `[]` if you want to run once, or remove the problematic dependency.

06 ■ AI Prompting Quick Reference

Copy and adapt these prompt templates for your homework and projects:

Debugging Error:

```
"I'm using React + Vite. I'm getting this error: [paste error]. Here's my component: [paste code].  
What is causing this and how do I fix it? Explain why it happened."
```

Understanding Code:

```
"Explain this React code line by line, as if I'm a beginner who has been learning React for 2 weeks:  
[paste code]"
```

Code Review:

```
"Review this React component. Tell me: Is there anything wrong? Is there a better way? Am I handling  
errors correctly? [paste code]"
```

Concept Explanation:

```
"Explain [concept] using a simple everyday analogy. No jargon. Like I'm 15."
```

API Help:

"I'm fetching from this API in React: [URL]. The response looks like this: [paste JSON]. How do I access the fields I need and display them in JSX? Show me example code."

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